Mobile-Augmented Smart Queue Management System for Hospitals

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***Abstract*—Management of high patient loads at tertiary hospitals presents a significant challenge in streamlining healthcare service delivery. Patients often need to queue up at various service areas in hospitals such as at registration, laboratory test and bill payment counters. Queue Management Systems (QMS) present a viable solution for patient management in such scenarios. However, conventional QMS are not generic by design, depend on hardware components or do not provide a comprehensive end-to-end solution that caters to the complete patient workflow. This paper presents a mobile-augmented smart queue management system that can be easily configured with an operational Hospital Management Information System (HMIS). It provides multiple interfaces for token generation and consumption on mobile devices integrated with hospital service counters, while using smart algorithms for token generation and allocation. The solution is comprehensive as that it caters to streamlined queue management across multiple hospital service areas using a single token for a patient, which improves patient experience and also helps the hospital administration track and optimize key performance metrics. We present the architectural and operational design of this system, along with an illustration of the use of this system in tracking productivity of service counter operators in a pilot implementation.**

***Index Terms*—Medical Informatics, Hospital Management Systems, Queue Management Systems, Token Generation Systems, Hospital Service Areas, Patient Counters**

I. Introduction

Hospital Management Information Systems (HMIS) typically comprise of a set of cohesive modules to cater to clinical and non-clinical workflows at a hospital. The interfaces for transaction management in HMIS includes various service areas such as patient registration counters, bill payment counters, laboratory test requisition, sample collection or report collection counters, pharmacy counters, as well as patient waiting areas for out-patient visits and appointments. Management of high patient loads in such areas is a challenge and efficient systems are needed, especially at tertiary care facilities. With increasing availability of mobile devices and ubiquitous technologies, the need for a smart and streamlined Queue Management System (QMS) has become imperative for improving patient experience and optimizing performance metrics for hospitals for streamlined healthcare service delivery.

There have been various works in literature pertaining to queue management for patients at hospitals for availing healthcare services. Hedau *et al.* [1] demonstrate the design and development of a queue management system for patients where an Android app provides interfaces to book appointments with doctors and notifications are sent to the patient in order to reduce waiting time at the OPD/doctor visiting area. Their app additionally provides navigation directions to the hospital. Aizan *et al.* [2] present a "walk-away" queue management system where the conventional token dispensers and token calling hardware at counters are replaced by Android devices and a service-based model for generation and management of tokens is presented. Notifications are sent to users at a specified time limit prior to anticipated calling of the token, which is enabled as a result of capturing the user's mobile number at the time of token generation. Sahney [3] enumerates the benefits and use cases of a smart queue management system at a prominent hospital in Delhi, India. Batbagon *et al.* [4] present the *IQueue* system with similar objectives and functionality, using a web-based QMS application supplemented by Android apps, along with supplementary services such as report generation and data analytics for optimizing queue generation at service areas. GSM based queue management systems have been presented in by Arun *et al.* [5] which integrates a PC based system with a microcontroller. Approaches using wireless technologies [6] and Internet of Things [7] have also been proposed. There have also been directions in literature that illustrate the benefits of employing a queue management system in improvement of hospital performance parameters such as patient length of stay [8], waiting time for in-patient surgeries [9], triaging in emergency departments [10], [11] and decision support [12].

The QMS proposed in this paper is intended to provide service-area based queue generation and management at hospitals, and would be generic in design for easy integration with an operational HMIS. An overview of the features of the QMS is shown in Fig. 1. This would enable the use of the system for token generation and queue management at multiple service areas such as patient registration counters, billing counters as well as hospital laboratories, pharmacy and

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DOI 10.1109/CBMS49503.2020.00086

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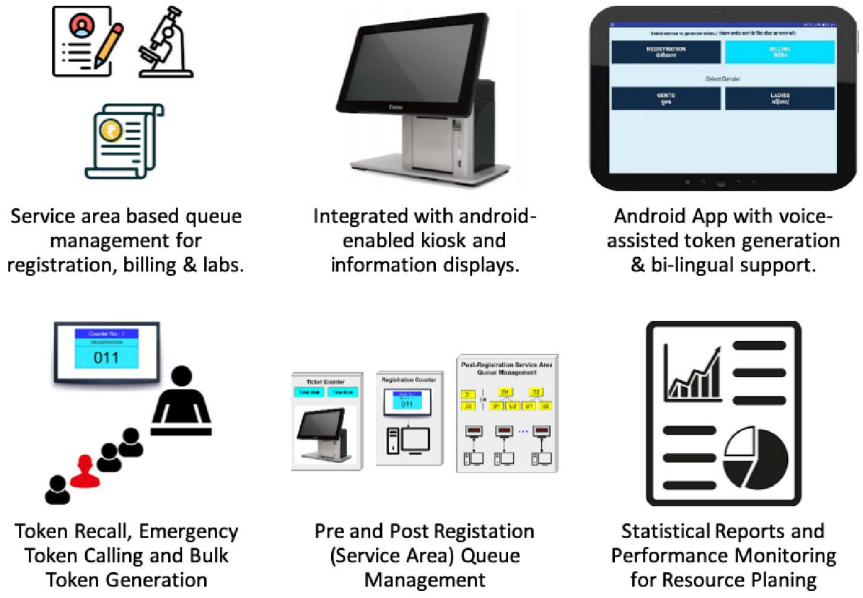


Fig. 1: Overview of the Mobile-Augmented QMS

patient waiting areas. The system is designed with interfaces on mobile devices and smart displays which makes it suitable for use without customized hardware support. The novelty in the proposed system includes crowd based token generation in multiple counters, where tokens are generated based on the number of tokens serviced at that counter. This would help in crowd control in high rush scenarios, which is common at government medical facilities in India. In addition, the system is also expected to support multiple token generation at a single counter, which could be used for cases where family members avail healthcare services collectively. The system also supports seamless queue management between various service areas (registration and doctor visit for example) by transferring the queue number from the registration desk to the OPD area, as well as for subsequent services, so that the patient need not generate/collect multiple tokens. The system would also enable the use of analytics on token generation data for optimizing patient waiting times and streamline the availability of hospital services.

The rest of this paper is organized as follows. Section II presents the detailed architecture, working and functional details of the proposed QMS. This is followed by results from a pilot implementation in Section III at a tertiary care facility where the QMS has been deployed and used for performance monitoring. Finally, conclusions and future work are presented in Section IV.

II. Mobile-Augmented QMS

We propose a mobile-augmented comprehensive QMS that can be used to handle diverse service-based workflows in hospitals. Our proposed system covers the complete patient workflow from token generation at registration to token consumption at multiple service areas depending on the requirements of the patient. This may include clinical consultation with doctors, lab tests requisition, sample collection or report collection, payment for healthcare services such as pharmacy, among others. The patient is provided with an Android based application that can be installed on the mobile devices as well as electronic kiosks available at the hospital. The patient uses the interface to generate a token for the services required on a particular day. Once the patient has generated the token, it is transferred to the various workflow counters and the patient receives relevant alerts or notifications when the token is called. All the consumption counters are provided with smart displays that interface with the system over secure services for updating the token status. Further, responsive interfaces are provided to all counter operators to call and consume the respective tokens. The comprehensive solution offers convenience to the patients, while hospital administration can monitor and plan resource deployment efficiently for service counters with an objective of increasing operational efficiency.

A. System Architecture

The system-level architecture of the QMS is shown in Fig. 2. There are three primary components of the proposed system. The first component responsible for token generation for queues is done using an Android based application which can be installed on the mobile device of the patient or on Android enabled electronic kiosks in the hospital premises. The app allows for category and service based token generation for multiple counters, with support for smart algorithms for token generation. The second component is the information display infrastructure which essentially comprises of a set of services called over a socket-based infrastructure that can be rendered in web pages on a browser which runs on smart TVs. These display units are patient information displays maintained at the

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respective service counters. The third component is the token consumption interfaces are provided which are integrated with the HMIS modules for the respective services for which QMS is configured and deployed.

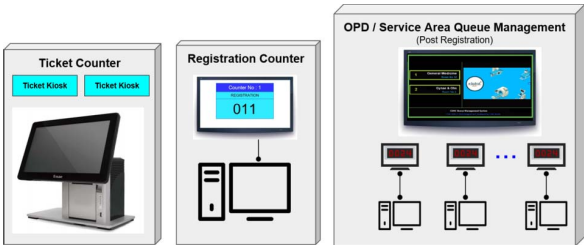


Fig. 2: Architecture of the QMS

B. Working of the Mobile-Augmented Smart QMS

The steps involved in the workflow of QMS are as follows:

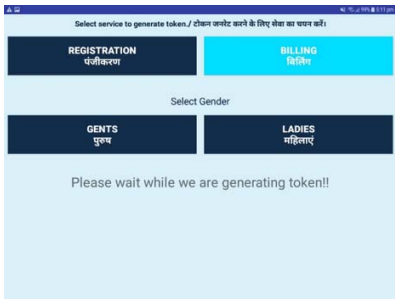
1) The patient can generate tokens at the kiosks. The tokens can be generated based on:
   a) Service Type (Registration/ Billing/ Investigation etc.)
   b) Patient Category (Gender/ Senior Citizen etc.)
   c) Hospital Organization (Department/ Unit wise)
2) After issue of token patient can wait at waiting area of respective counter. Information display boards would be provided at this counter which will provide details of the token being served at the respective counter.
3) At the service counter, the counter operator has an interface to call the next token generated in the queue. Once the counter operator calls the next token, patient visits the counter with the token generated from the kiosk and avails the required service.
4) The token status is updated in the system back-end. This also allows generation of performance reports and monitoring operational parameters of the QMS.

In the following subsection we discuss in detail the features and algorithm being used in the QMS for efficient management of patient queues.

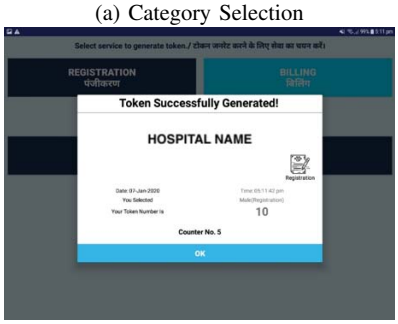
C. Features of the Mobile-Augmented Smart QMS

The key features of the Mobile-Augmented smart QMS are as follows:

1) Configuration of service-based counters by administrator: This facility would allow the QMS administrator to configure the number of counters available for generation of queues based on the service area type, for eg. in case of services such as registration, investigation and billing, number of counters available for management of patient queues can be configured. This would also enable mapping of users to manage counters, disabling/enabling counter operation for servicing tokens, and creation and management of new counters.



(a) Category Selection



(b) Token Preview

Fig. 3: Token Generation Interfaces

2) Generation and issue of tokens for queue management at service areas: This facility allows patient to generate tokens for waiting in queues at counters for the respective service areas. The interfaces for token generation are provided on mobile-based or kiosk-based platforms that can be easily accessed by users. The token generation typically collects basic demographic details of patients and issues a token number for the required service type. Mobile number of the user is captured for providing alerts/notifications for patients on mobile. Representative interfaces for token generation are shown in Fig. 3, where the screen for choosing the patient category and service type for which token is required is shown in Fig. 3a, while the generated token is shown in Fig. 3b.

3) Smart Queue Generation: In order to streamline large-scale patient queue management, dynamic queue generation has been provided which generates queue numbers based on the crowd at the respective counters, where the next queue number would be assigned to the counter having lower number of pending tokens to call.

The algorithm for generation of tokens based on this approach is summarized in Algorithm 1. The approach takes a list of counters C along with their criteria for patient admission specified by the set S. This may include restrictions such as counters for females, senior citizens, hospital staff, specific insurance beneficiaries, among others. In addition, the allocation lists of the counters are maintained in L. The approach returns a counter C_P and corresponding token/queue number T_P for a patient P. This is done by first identifying the set of eligible counters C_{sat} which can be potentially allocated to the patient. An indicator function $f(\cdot)$ checks the patient

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criteria with the counter satisfiability criteria and returns 1 for the set of counters that match. Once the set of eligible counters have been identified, the counter with the least number of allocated tokens is identified and the next available token number is generated and returned to the patient. In case all counters have equal or no allocation, an eligible counter is chosen at random and the next token is assigned to the patient. In both cases, the corresponding counter allocation lists are updated.

Data: Counters $(C_1, C_2, ..., C_N)$, Allocation Lists $(L_1, L_2, ..., L_N)$, Satisfying Set $(S_1, S_2, ..., S_N)$, New Patient Criteria (P)
Result: Counter (C_P) and Token (T_P) for Patient P
// Find counters that can be allocated to patient.
Initialize $C_{sat} \leftarrow \{\phi\}$;
for *i=1:N* **do**
  if $f(S_i, P) == 1$ // $f(\cdot) = 1$ if eligible **then**
    $C_{sat} \leftarrow \{C_{sat} | C_i\}$;
  end
end
// Find counter with minimum allocation
$C_{min} \leftarrow \min_i \{length(L_i) | i \in C_{sat}\}$;
// Determine counter
if $C_{min} \neq \{\phi\}$ **then**
  $C_P \leftarrow C_{min}$;
else
  // Allocate any eligible counter
  $C_P \leftarrow C_{rand(i)} | C_i \in C_{sat}$;
end
// Get token number
Token $T_P \leftarrow L_P(C_P) + 1$;
// Update allocation list
Update $L_P \leftarrow \{L_P | T_P\}$;
Return $\langle C_P, T_P \rangle$;

Algorithm 1: Crowd-based dynamic token generation algorithm for smart QMS.

4) Patient Wait Time Estimation: Once adequate transactional data has been collected by the system for token generation at counters, the QMS would dynamically estimate patient wait time based on past trends and the same would be sent as a notification to the patient. This would enable the patient to accordingly plan the hospital visit and avail the service required.

5) Token Skip and Recall: This feature allows the counter operator to skip the next token in sequence or call an out-of-sequence token. The counter display screen gets updated accordingly. This feature can be used for situations where manual intervention for token servicing (by violation of queue sequence) is required.

6) Family based Token Generation: In cases where multiple members of the same family avail a service (eg. patient registration), the system allows generation of multiple token numbers assigned to the same counter for the convenience of the user. This is also beneficial for patient visits to multiple departments, where a multiple tokens can be generated at the same counter, and patient can request for multiple registrations/visit stamping at hospitals for individual departments.



(a) Single Counter



(b) All Counters

Fig. 4: Counter-based information displays for QMS

7) Integrated Counter Displays: This facility includes services for fetching details of counter-wise generated token numbers that will be displayed on information screens at the respective counters. This would help patients identify the ongoing token number being served and the next token number that would be called at the specific counter. Fig. 4 shows simple interfaces for counter information displays, where the display at a specific counter is shown in Fig. 4a and a consolidated display of all operational counters for the set of service categories are shown in Fig. 4b.

8) Messaging and Notification Services: The QMS is also integrated with messaging and notification services where the patient would receive notifications on the mobile device when the tokens are generated at the kiosk or from the app. The expected waiting time is also intimated to the patient. Further, the patient is also sent a notification when the token is called at a specific service counter. The notifications can be configured over SMS or suitable cloud messaging services.

III. Operational Monitoring using QMS

One of the primary benefits of deployment of the mobile-augmented smart QMS system is the fact that the use of a dynamic token generation and allocation policy allows for monitoring the performance of the service counters. In order to illustrate the same, the developed system was deployed on a pilot basis at a tertiary care hospital where patient registration at five registration counters was managed through the QMS. The hospital has an operational HMIS [13], [14]

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with multiple modules and Electronic Health Record (EHR) standards compliance [15], [16], [17], [18].

Each counter was assigned to a specific operator who would login to the HMIS and use the QMS interfaces to consume the tokens generated by the patients from the registration kiosks or their mobile devices. Four of these counters did not have any service restrictions while the fifth counter had a service type condition for billing (it was used to attend to patient billing requests as opposed to the other counters which catered to patient registration requests). It may be noted here that the tokens consumed at the counters were carried on for further workflow, which usually included queue management at the doctor's consultation area. This was done department-wise and the entire patient workflow was managed using the single queue number generated prior to consumption at the registration or billing counters.

Table I shows the number of tokens that have been generated at each counter along with the number of tokens that were attended, skipped and unattended for each day for a period of five days. It may be noted here that the tokens were generated using the dynamic allocation algorithm discussed previously. The percentage of the number of attended tokens to the total number of tokens generated at a counter can be used as simple metric to determine the operational efficiency of a counter operator.

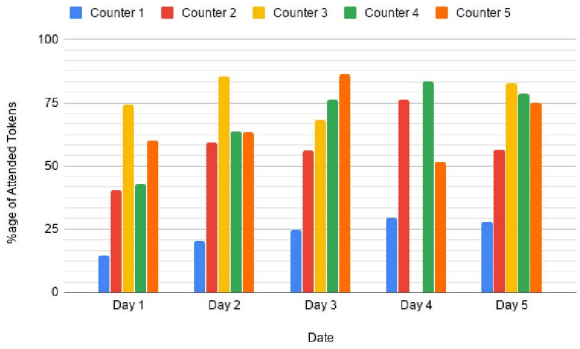


Fig. 5: Percentage token consumption across counters.

A plot of the percentage token consumption ratio at the respective counters is shown in Fig. 5. From the results obtained, it can be seen that the highest average token consumption is at Counter 5, while the lowest is at Counter 1. These results, aggregated over time, can be used to suitably incentivize high performing counter operators.

A. Hospital Key Performance Metrics

The QMS also provides an integrated dashboard view that computes Key Performance Indicators (KPIs) for hospitals from the transactional data. An illustration of the dashboard view is shown in Fig. 6.

The dashboard shows various KPIs including average time taken for patient from token generation to registration where the token is consumed, average time taken from registration



Fig. 6: Dashboard view for hospital KPIs (representative values).

to vitals recording, average time from vitals recording to consultation with the clinician, time taken for the consultation process itself, average time taken from lab test requisition to billing, average time from billing to sample collection for the lab test and average time from sample collection to lab test report generation. It may be noted here that since the QMS uses the same token for tracking the entire patient workflow, these parameters can be computed and monitored. Further, it is also easy for the patient as he need not take multiple tokens. These KPIs can be tracked over time using the QMS and can be monitored by the hospital administration for improving resource planning and allocation.

B. Trend Analysis and Prediction

The data obtained above can also be used for estimating the patient loads at hospitals based on past data. As a representative example, we fit polynomial curves of degree two to the tokens generated at each counter and show the trendlines in Fig. 7. It may be noted here that the sample size in terms of number of days for which the data is available is small, and a valid analysis mandates a larger sample size. However, the idea is to illustrate the potential analysis that can be done so that it can be adapted and implemented on scale at the hospitals.

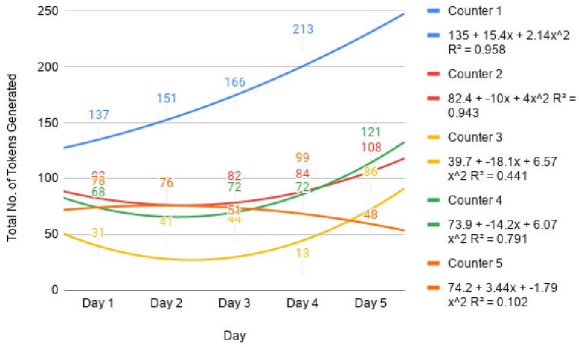


Fig. 7: Trendlines for tokens generated at counters.

Fig 7 shows the trendlines for the individual counters along with the second order model coefficients fitted for each counter. The R2 values in terms of goodness of fit are also

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TABLE I: Performance Monitoring using smart QMS

Day	All Counters				Counter 1			Counter 2			Counter 3			Counter 4			Counter 5		
	Tot	Att	Skip	Unatt	Att	Tot	%age	Att	Tot	%age	Att	Tot	%age	Att	Tot	%age	Att	Tot	%age
Day 1	396	152	238	6	20	137	14.6	33	82	40.24	23	31	74.19	29	68	42.65	47	78	60.26
Day 2	421	208	202	11	31	151	20.5	45	76	59.21	35	41	85.37	49	77	63.64	48	76	63.16
Day 3	415	216	178	21	41	166	24.7	46	82	56.10	30	44	68.18	55	72	76.39	44	51	86.27
Day 4	481	238	216	27	63	213	29.6	64	84	76.19	0	13	0.00	60	72	83.33	51	99	51.52
Day 5	589	326	241	22	63	226	27.9	61	108	56.48	71	86	82.56	95	121	78.51	36	48	75.00
Total	2302	1140	1075	87	Avg(%)		23.5	Avg(%)		57.64	Avg(%)		62.06	Avg(%)		68.90	Avg(%)		67.24
Tot = Total no. of tokens generated, Att = No. of tokens attended, Skip = No. of tokens skipped, Unatt = No. of tokens unattended																			

indicated against each model. It can be seen that the trend for tokens generated typically follows an increasing trend for all counters except counter 5.

Using the model coefficients determined for the individual counters, the projected patient load can be estimated. The confidence of such a prediction can be estimated by the R2 values [19] computed for each counter from the model fitted to the past data. For this specific example, it can be seen that the R2 value for counters 1 and 2 is closest to 1, hence its prediction could be associated with a higher confidence level. The lowest confidence level is for counter 5. Projected estimates using such models can be used to plan future resource allocation for streamlining patient experience at counters and eventually reduce waiting time.

IV. Conclusion and Future Work

This paper presented a mobile-augmented smart QMS that can be used for end-to-end patient management at hospitals. The developed system is generic for integration with operational HMIS and provides platform agnostic interfaces on mobiles, desktops and smart TVs for counter information displays. It incorporates smart queue generation algorithms for streamlining patient experience and managing the patient load at hospital service counters, while also supporting operational monitoring for resource planning and allocation at service areas by hospital administrators. Future work involves analysis of larger datasets from the pilot implementation and developing enhanced crowd management and resource requirement forecasting models for hospitals based on the QMS.

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